higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

MASONRY
NQF LEVEL 2

(12020152)

12 March 2019 (X-Paper)
09:00–12:00

Drawing instruments may be used.

This question paper consists of 7 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Make ALL drawings in the ANSWER BOOK.
5. Rule off across the page on completion of each question.
6. Write neatly and legibly.

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QUESTION 1: GENERAL

1.1 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.1.1–1.1.5) in the ANSWER BOOK.

1.1.1 A chalk line is a thin line attached to profiles to help keep courses straight.

1.1.2 The jambs of steel doorframes are the vertical sides of the doorframes.

1.1.3 As stipulated in the regulated national standards, the minimum period that lintels should be propped is three days.

1.1.4 A single-leaf wall has a single row of bricks laid end-to-end usually 110 mm thick.

1.1.5 The oldest brick building was found in Africa and built around 15 000 years ago.

(5 × 1) (5)

1.2 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.2.1–1.2.5) in the ANSWER BOOK.

1.2.1 Masonry units are joined together by using …

A concrete.
B mortar.
C cement.
D topping.

1.2.2 Mortar is a mixture of …

A sand, cement, stone and water.
B cement, stone and sand.
C cement, sand and water.
D lime and water.

1.2.3 A bond with stretchers and headers laid alternately in a single course:

A English bond
B Stretcher bond
C Header bond
D Flemish bond

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1.2.4 Corners of brickwork must be reinforced with brick force folded at …

A 30°.
B 45°.
C 60°.
D 90°.

1.2.5 Which type of joint is regarded as poor according to the CMA?

A Concave joints
B Struck joints
C Weathered joints
D Flush joints

(5 × 1) (5)
[10]

QUESTION 2: INTRODUCTION AND BACKGROUND

2.1 A cavity wall consists of two leaves of brickwork built parallel to each other with a space between the leaves. The two leaves are tied together by wall ties.

List THREE types of wall ties that can be used. (3)

2.2 A cavity wall is built with a 50 mm space between the two leaves. Each leaf is 110 mm thick.

Calculate the overall thickness of the cavity wall. (1)

2.3 Define each of the following terms:

2.3.1 Hollow masonry unit

2.3.2 Solid masonry unit

(2 × 2) (4)
[8]

QUESTION 3: SETTING OUT MASONRY WALLS

3.1 The corner of a building must be at a right angle when setting out.

Explain how the 3:4:5 method can ensure that the corner of a building is at a right angle. (5)

3.2 Explain the steps to follow when setting out and marking a profile. (5)

3.3 Screed rails are used when making concrete floors or concrete slabs.

3.3.1 Give TWO advantages of using screed rails.

3.3.2 State TWO ways to maintain screed rails when removed.

(2 × 2) (4)

3.4 The correct tools must be used to complete construction-project activities correctly.

Name the correct tool for each of the following activities:

3.4.1 Smooth the mortar joints between bricks

3.4.2 Check if corners of walls are square

3.4.3 Apply mortar upon the base on which bricks or blocks can be laid

3.4.4 Thin line attached to profiles to help keep courses straight

3.4.5 Scratch coat plaster before beginning the second layer of plaster

3.4.6 Mark straight lines by striking the line on foundations

(6 × 1) (6)

3.5 A gauge rod is made up by taking a variance of mortar thickness including the thickness of each brick marked on the gauge rod which is then transferred to the profiles to ensure the wall is level upon completion. This gauge rod cannot be used on every building site.

Give THREE reasons why the same gauge cannot be used on other building sites. (3)

[23]

QUESTION 4: BASIC CONSTRUCTION OF MASONRY WALLS

4.1 There are various types of bonds that can be used to construct a wall. How a brick is laid will determine the type of bond.

Draw the front view of each of the following types of bonds (NO stopped ends required):

4.1.1 Stretcher bond

4.1.2 English bond

4.1.3 Flemish bond

(3 × 4) (12)

4.2 A damp-proof course can be used in various ways and the installation of damp-proof courses must be in accordance with the drawings and provisions of national standards.

4.2.1 Explain the procedure to lay down horizontal DPC. (6)

4.2.2 Give THREE reasons why DPC should not be used on a free-standing garden wall. (3)

4.3 Explain why clay masonry units and concrete masonry units may not be randomly mixed on a wall. (3)

4.4 Explain the circumstances where clay masonry units may be used with concrete masonry units. (3)

4.5 Define the construction term *unit*. (1)

[28]

QUESTION 5: ADVANCED CONSTRUCTION OF MASONRY WALLS

5.1 Define each of the following masonry terms:

5.1.1 Roof anchor

5.1.2 Brick arch

5.1.3 Keystone

5.1.4 Control joint

5.1.5 Composite beam

5.1.6 Prestressed concrete lintels

5.1.7 Gauge rod

(7 × 2) (14)

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5.2 Explain how to correctly install a prestressed lintel. (6)

5.3 To cure a lintel means to give the lintel time to set and obtain its full strength over a period of days keeping it moist.

Give TWO things that could happen if lintels are not allowed to cure. (2)

5.4 Explain the difference between *racking* and *toothing*. (2 + 2) (4)

5.5 List FOUR materials used to make concrete. (4)

5.6 Give the function of roof anchors. (1)

[31]

TOTAL: 100

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